1. data model in **C# with EF Core 8**:

---

1. Define the Entity Classes

```
using System.Collections.Generic;

namespace UniversityEFCore.Models
{
    public class Department
    {
        public int Id { get; set; }
        public string Name { get; set; }
        public string MgrName { get; set; }

        // Navigation properties
        public ICollection<Student> Students { get; set; }
        public ICollection<Teacher> Teachers { get; set; }
        public ICollection<Course> Courses { get; set; }
    }

    public class Teacher
    {
        public int Id { get; set; }
        public string Name { get; set; }
        public decimal Salary { get; set; }
        public string Address { get; set; }

        // Foreign Keys
        public int CourseId { get; set; }
        public int DepartmentId { get; set; }

        // Navigation properties
        public Course Course { get; set; }
        public Department Department { get; set; }
    }

    public class Course
    {
        public int Id { get; set; }
        public string Name { get; set; }
        public int Degree { get; set; }
        public int MinDegree { get; set; }
        public int DepartmentId { get; set; }

        // Navigation properties
        public Department Department { get; set; }
        public ICollection<Teacher> Teachers { get; set; }
        public ICollection<StuCrsRes> StuCrsRes { get; set; }
    }

    public class Student
    {
        public int Id { get; set; }
        public string Name { get; set; }
        public int Age { get; set; }
        public int DepartmentId { get; set; }

        // Navigation properties
```

```
        public Department Department { get; set; }
        public ICollection<StuCrsRes> StuCrsRes { get; set; }
    }

    public class StuCrsRes
    {
        public int StudentId { get; set; }
        public int CourseId { get; set; }
        public int Grade { get; set; }

        // Navigation properties
        public Student Student { get; set; }
        public Course Course { get; set; }
    }
}
```

---

2. Define the `DbContext`

```
using Microsoft.EntityFrameworkCore;

namespace UniversityEFCore.Models
{
    public class UniversityContext : DbContext
    {
        public UniversityContext(DbContextOptions<UniversityContext>
options) : base(options) { }

        public DbSet<Department> Departments { get; set; }
        public DbSet<Teacher> Teachers { get; set; }
        public DbSet<Course> Courses { get; set; }
        public DbSet<Student> Students { get; set; }
        public DbSet<StuCrsRes> StuCrsRes { get; set; }

        protected override void OnModelCreating(ModelBuilder modelBuilder)
        {
            // Composite Key for StuCrsRes
            modelBuilder.Entity<StuCrsRes>()
                .HasKey(scr => new { scr.StudentId, scr.CourseId });

            // Relationships
            modelBuilder.Entity<Department>()
                .HasMany(d => d.Students)
                .WithOne(s => s.Department)
                .HasForeignKey(s => s.DepartmentId);

            modelBuilder.Entity<Department>()
                .HasMany(d => d.Teachers)
                .WithOne(t => t.Department)
                .HasForeignKey(t => t.DepartmentId);

            modelBuilder.Entity<Department>()
                .HasMany(d => d.Courses)
                .WithOne(c => c.Department)
                .HasForeignKey(c => c.DepartmentId);

            modelBuilder.Entity<Teacher>()
                .HasOne(t => t.Course)
                .WithMany(c => c.Teachers)
                .HasForeignKey(t => t.CourseId);
```

```
            modelBuilder.Entity<StuCrsRes>()
                .HasOne(scr => scr.Student)
                .WithMany(s => s.StuCrsRes)
                .HasForeignKey(scr => scr.StudentId);

            modelBuilder.Entity<StuCrsRes>()
                .HasOne(scr => scr.Course)
                .WithMany(c => c.StuCrsRes)
                .HasForeignKey(scr => scr.CourseId);
        }
    }
}
```

---

3. Apply Migration and Update Database

Run these commands in **Package Manager Console**:

```
Add-Migration InitialCreate
Update-Database
```

This will create all the tables (`Departments`, `Teachers`, `Courses`, `Students`, `StuCrsRes`)

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two MVC pages:

1. **ShowAll** → `/student/ShowAll` → List all students.
2. **ShowDetails** → `/student/ShowDetails?id=3` → Show details of one student by Id.

Here's how you can implement this in **ASP.NET Core MVC with EF Core 8**:

---

1. StudentController

```
using Microsoft.AspNetCore.Mvc;
using Microsoft.EntityFrameworkCore;
using UniversityEFCore.Models;
using System.Threading.Tasks;
using System.Linq;

namespace UniversityEFCore.Controllers
{
    public class StudentController : Controller
    {
        private readonly UniversityContext _context;

        public StudentController(UniversityContext context)
        {
            _context = context;
        }

        // /student/ShowAll
        public async Task<IActionResult> ShowAll()
        {
            var students = await _context.Students
```

```
                .Include(s => s.Department)
                .ToListAsync();

            return View(students);
        }

        // /student/ShowDetails?id=3
        public async Task<IActionResult> ShowDetails(int id)
        {
            var student = await _context.Students
                .Include(s => s.Department)
                .Include(s => s.StuCrsRes)
                    .ThenInclude(scr => scr.Course)
                .FirstOrDefaultAsync(s => s.Id == id);

            if (student == null)
            {
                return NotFound();
            }

            return View(student);
        }
    }
}
```

---

2. Views

Views/Student/ShowAll.cshtml

```
@model IEnumerable<UniversityEFCore.Models.Student>

<h2>All Students</h2>
<table class="table table-bordered">
    <thead>
        <tr>
            <th>Id</th>
            <th>Name</th>
            <th>Age</th>
            <th>Department</th>
            <th>Actions</th>
        </tr>
    </thead>
    <tbody>
    @foreach (var student in Model)
    {
        <tr>
            <td>@student.Id</td>
            <td>@student.Name</td>
            <td>@student.Age</td>
            <td>@student.Department?.Name</td>
            <td>
                <a href="/student/ShowDetails?id=@student.Id" class="btn
btn-info">Details</a>
            </td>
        </tr>
    }
    </tbody>
</table>
```

Views/Student/ShowDetails.cshtml

```
@model UniversityEFCore.Models.Student

<h2>Student Details</h2>

<div>
    <strong>Id:</strong> @Model.Id <br />
    <strong>Name:</strong> @Model.Name <br />
    <strong>Age:</strong> @Model.Age <br />
    <strong>Department:</strong> @Model.Department?.Name <br />
</div>

<h3>Courses & Grades</h3>
<table class="table table-striped">
    <thead>
        <tr>
            <th>Course</th>
            <th>Grade</th>
        </tr>
    </thead>
    <tbody>
    @foreach (var res in Model.StuCrsRes)
    {
        <tr>
            <td>@res.Course.Name</td>
            <td>@res.Grade</td>
        </tr>
    }
    </tbody>
</table>

<a href="/student/ShowAll" class="btn btn-secondary">Back to List</a>
```